Preliminary Project Report On

Lung Cancer Detection using Deep Learning

Submitted in partial fulfilment of the
requirements for the Degree of

BACHELOR OF ENGINEERING
in
INFORMATION TECHNOLOGY

By

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Certificate

This is to certify that following students

This is to certify that the Dissertation Project Stage I report entitled

Lung Cancer Detection using Deep Learning

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is a bonafide work carried out by them under the supervision of Prof. K. S. Bhagwat and it is submitted towards the partial fulfillment of the requirement ofSavitribai Phule Pune University, Pune for the award of the degree of Bachelor of Engineering.

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1 Abstract

In terms of the number of deaths caused by cancer worldwide, lung cancer is the most common cause. CT scans are used by radiologists to detect and monitor cancer in the human body. It is possible that the visual interpretation of a database may lead to the detection of cancer at an advanced stage, which only serves to increase cancer death rates. Because of this, image processing can be used to detect cancer at an early stage. Using mathematical morphological operations, this paper proposes a lung cancer detection algorithm that extracts Haralick features and uses them to classify cancer using artificial neural networks after segmenting the region of interest in the lung.

Here, we design a new convolutional neural network for this purpose. Marine predators algorithm is also used for optimal arrangement and better network accuracy. The method finally applied to RIDER dataset, and the results are compared with some pretrained deep networks, including CNN ResNet-18, GoogLeNet, AlexNet, and VGG-19. Final results showed higher results of the proposed method toward the compared techniques. The results showed that the proposed MPA-based method with 93.4% accuracy, 98.4% sensitivity, and 97.1% specificity provides the highest efficiency with the least error (1.6) toward the other state of the art methods.

2 Technical Words

1. Computer tomography
2. Lung cancer
3. Features
4. Neural networks.

3 Introduction

3.1 Introduction

Lung cancer disease is the second largest death threat to the world after heart attack, as this cancer is responsible for the largest number of deaths, compared to the number of deaths caused by any other cancer type. [1]. Lung cancer is the uncontrolled growth of the cells, thus leading to the formation of lung nodules. It is reported that lung cancer is responsible for around 19% deaths globally mostly due to alcohol and tobacco consumption. The rate of survival is assured by only 15% survival chances, for a survival period of 5 years. [2]. The main cause of such high death rate is the detection in later stages, thus leading to delayed treatment. If lung cancer is detected at an earlier stage, chances of survival can increase up to 50-70%. Non small cell lung cancer and small cell lung cancer are the two major groups into which the lung cancer can be classified based on the cell characteristics. [7] Non small cell lung cancer is the most common type of lung cancer contributing to about 85-90% of total lung cancer cases, while the other 10-15% of the cases is diagnosed with small cell lung cancer. [3] The extent of the spread of cancer is the basis for the division of lung cancer into stages. It comprises of four stages namely stage I-The cancer is confined to the lung, stages II and III-the cancer is confined to the chest (with larger and more invasive tumors classified as stage III) and Stage IV-Cancer has spread from the chest to other parts of the body. There are many techniques to diagnose the lung cancer such as X-rays, Computed Tomography (CT), Magnetic Resonance Imaging (MRI scan), and Sputum Cytology. The problem with these techniques is that it can be time consuming and makes detection possible only at later stages[2].

Among lung diseases, lung cancer is still recognized as one of the most dangerous cancers. One-third of cancer deaths are due to lung cancer. About 80% of patients remain in the best condition for five years after being diagnosed with this type of cancer. Air pollution is one of the main causes of this disease. Early diagnosis of lung disease will have a major impact on the likelihood of definitive treatment of the disease. Major diagnostic methods for lung cancer include imaging, radiography and CT scan, biopsy, bronchoscopy, and examination of the breast mucosa. Pulmonary nodule is a small, round, opaque mass that forms inside the lung tissue. In other words, nodules are spherical radiographic opacities less than 30 mm in diameter. So far, various researches have been done in order to identify and describe lung diseases. Due to the removal and high number of radiographic images of the lungs, as well as its complex and uneven structure, it is difficult for a specialist physician to diagnose nodules from vessels, wounds, etc. A computer diagnostic aid system is a system that assists a physician in diagnosing a disease. Such systems are used as an intelligent tool that expresses a second option to the radiologist, which shows suspicious situations in the images to the radiologist and thus helps the radiologist to make the most accurate diagnosis. The basic idea is not to leave the diagnosis to a machine, but to use it because it increases the sensitivity of the work and reduces the rate of positive error. Several works were introduced in this subject.

3.2 Motivation

- Cancer is the root cause for a large number of deaths worldwide, out of which lung cancer is the cause of the highest mortality rates.
- Computer tomography scan is employed by radiologists to detect cancer in the body and track its growth.
- Visual interpretation of database can lead to cancer detection at later stages, thus leading to late treatment of cancer which only boosts up the cancer death rates.

4 Literature Survey

Literature survey is the most important step in any kind of research. Before start developing we need to study the previous papers of our domain which we are working and on the basis of study we can predict or generate the drawback and start working with the reference of previous papers.
“In this section, we briefly review the related work on Lung Cancer detection system and their different techniques.

In the last few years Convolutions neural networks have become dominant approach in medical image analysis that outperforms algorithms based on handcrafted features. Esteva et al. [6] proposed the CNN-based system for skin lesion classification that achieves performance comparable with dermatologist’s assessment. Regarding lung cancer diagnosis, methods proposed so far have dealt mostly with radiology. In [7] image-based radiomics features strongly related to survival are extracted from positron emission tomography-computed tomography (PET/CT) scans. In [8] CNN is employed for classification of lung nodule images yielding accuracy of 86.4%. In digital pathology tasks CNNs chave been used on cell level for mitosis detection [9] and cell nuclei detection [10]. CAMELYON16 was the first challenge dealing with WSIs to detect breast cancer metastases in lymph nodes. Thanks to availability of large annotated training set in this challenge, it was possible to train deeper and more powerful CNN architectures like GoogLeNet [11], VGG-Net [12] and ResNet [13]. Method that gives best result in this challenge is described in [14]. It performs patch-based classification to discriminate tumor patches from normal patches using combination of 2 GoogLeNet architectures where one of them is trained with and another without hard-negative mining. Aim of TUPAC challenge was WSI based mitosis detection in breast cancer tissue and tumor grading prediction. In best performing method [15] ROI regions are firstly extracted from WSI based on cell density. This is followed by mitosis detection using ResNet CNN architecture. Finally, each WSI is represented by feature vector including the number of mitoses and cells in each patch as well as other features derived from statistics. This feature vector is fed to SVM classifier to predict tumor proliferation.

5 Research Methodology

5.1 Convolution Neural Network(CNN)

The structure of CNN includes two layers one is feature extraction layer, the input of each neuron is connected to the local ready fields of the previous layer, and extracts the local feature. Once the local features are extracted, the positional relationship between it and other features also will be displayed. The other is feature map layer; each computing layer of the network is collected of an advantage of feature map. Every feature map is a plane, the weight of the neurons in the plane are same. The structure of feature plan uses the sigmoid function as activation function of the convolution network, which makes the feature map have shift in difference. Besides, since the neurons in the same mapping plane share weight, the number of free parameters of the network is decreased. Each convolution layer in the convolution neural network is come after by a computing layer which is used to find the local average and the second extract; this unique two feature extraction structure decreases the resolution.

1. Convolutional Layer This layer is the first layer that is used to extract the various features from the input images. In this layer, the mathematical operation of convolution is performed between the input image and a filter of a particular size MxM. By sliding the filter over the input image, the dot product is taken between the filter and the parts of the input image with respect to the size of the filter (MxM). The output is termed as the Feature map which gives us information about the image such as the corners and edges. Later, this feature map is fed to other layers to learn several other features of the input image.

2. Pooling Layer In most cases, a Convolutional Layer is followed by a Pooling Layer. The primary aim of this layer is to decrease the size of the convolved feature map to reduce the computational costs. This is performed by decreasing the connections between layers and independently operates on each feature map. Depending upon method used, there are several types of Pooling operations.

3. Fully Connected Layer The Fully Connected (FC) layer consists of the weights and biases along with the neurons and is used to connect the neurons between two different layers. These layers are usually placed before the output layer and form the last few layers of a CNN Architecture.

Step1: Select the dataset.

Step2: Feature selection using information gain and ranking

Step3: Classification algorithm

Step4: Each Feature calculate fx value of input layer

Step5: bias class of each feature calculate

Step6: Next produce the feature map it goes to forward pass input layer

Step7: Calculate the convolution cores in a feature pattern

Step8: Produce sub sample layer and feature value.

Step9: Back propagation input deviation of the kth neuron in output layer.

Step10: Finally give the selected feature and classification results.

6 Proposed Work

6.1 Problem Definition

To design a comfortable and convenient Lung Cancer Detection System using Deep L-earning for faster and accurate results,to avoid malignancy score.

6.2 Scope of Project

Early diagnosis of lung cancer is critical for improvement of patient survival. Histopathological assessment of tissue is standard procedure needed for early diagnosis. Tissue analysis is usually performed by pathologist review, but this procedure is time-consuming and error-prone. Automated detection of cancer regions would significantly speed up the whole process and help the pathologist. In this paper we propose fully automatic method for lung cancer detection in whole slide images of lung tissue samples. Classification is performed on image patch level using convolutional neural network (CNN).

6.3 Project Objectives

- To implement Segmentation of cancer lesions from CT scans of COVID-19 patients for accurate diagnosis and follow-up.
- To propose a novel noise-robust framework to learn from noisy labels for the segmentation
- To improve accurate diagnosis using deep learning

7 Project Requirement Specification

7.1 Performance Requirements

The performance of the system lies in the way it is handled. Every user must be given proper guidance regarding how to use the system. The other factor which affects the performance is the absence of any of the suggested requirements.

7.2 Software Quality Attributes

7.2.1 Accuracy: -

The level of accuracy in the proposed system will be higher. All operation would be done correctly and it ensures that whatever information is coming from the center is accurate.

7.2.2 Reliability: -

The reliability of the proposed system will be high due to the above stated reasons. The reason for the increased reliability of the system is that now there would be proper storage of information.

7.2.3 No Redundancy: -

The main objective of proposed system is to prevent data from duplication on server. This would assure economic use of storage space and consistency in the data stored.

7.2.4 Immediate storage of information: -

In manual system there are many problems to store the largest amount of information.

7.2.5 Easy to Operate: -

The system should be easy to operate and should be such that it can be developed within a short period of time and fit in the limited budget of the user

7.3 Safety Requirements

To ensure the safety of the system, perform regular monitoring of the system so as to trace the proper working of the system. An authenticated user is only able to access system.

7.4 Security Requirements

Any unauthorized user should be prevented from accessing the system. Password authentication can be introduced.

7.5 Software Requirements

- Operating System - Windows 7/8/10
- Application Server - Apache Tomcat 7/8/9
- Front End - HTML, JDK 1.8, JSP
- Scripts - JavaScript.
- Server side Script - Java Server Pages.
- Database - My SQL 5.0
- IDE - Eclipse Oxygen
- Tool - Weka

7.6 Hardware Requirements

- Processor - Intel i3/i5/i7
- Speed - 1.1 GHz
- RAM - 2 GB(min)
- Hard Disk - 40 GB
- Floppy Drive - 1.44 MB
- Key Board - Standard Windows Keyboard
- Mouse - Two or Three Button Mouse
- Monitor - SVGA

8 Project Planning

8.1 Project Estimates

8.1.1 Cost Estimate

Cost will estimate after completing the project that depend on time to complete the project. Also efforts required to complete.

8.1.2 Time Estimates

Time will depend on modules of project.Also project plan of execution.

8.1.3 Project Resources

1. Hardware Resources Required
   System: Pentium IV 2.4 GHz. Hard Disk: 40 GB. Floppy Drive: 44 Mb. Monitor: 15 VGA Color.

2. Software Resources Required
   Operating system: Windows. Coding Language: Java 1.8 Database: MySql 5 IDE: Eclipse

8.2 Risk Management

8.2.1 Risk Identification

For risks identification, review of scope document, requirements specifications and schedule is done. Answers to questionnaire revealed some risks. Each risk is categorized as per the categories mentioned in [**?**]. Please refer table **??** for all the risks. You can refereed following risk identification questionnaire.

1. Have top software and customer managers formally committed to support the project?
Answer: Yes , Have top software and customer managers formally committed to support the project
2. Are end-users enthusiastically committed to the project and the system/product to be built?
Yse, end-users enthusiastically committed to the project and the system/product to be built
3. Are requirements fully understood by the software engineering team and its customers?
Yes, Are requirements fully understood by the software engineering team and its customers
5. Have customers been involved fully in the definition of requirements?
Yes, customers been involved fully in the definition of requirements
6. Are project requirements stable?
Answer: all project requirements are stable
9. Is the number of people on the project team adequate to do the job?

Yse, the number of people on the project team adequate to do the job

8.2.2 Risk Analysis

DESCRIPTION	lOW	High
Login detail	no	yes
Internet connection slow	yes	no

Table 1: Risk Analysis

Following are the details for each risk.
Risk ID 1
Risk Description Description 1
1.While login to system validation is there so user must follow rules and propoer enteries
Risk ID 2
1.While registering to system internet connection should be their for registration to get users data online

8.3 Project Schedule

8.3.1 Project task set

Major Tasks in the Project stages are:

- Task 1: Requirement Analysis (Base Paper Explanation).
- Task 2: Project Specification (Paper Work).
- Task 3: Technology Study and Design.
- Task 4: Coding and Implementation (Module Development).

8.3.2 Task network

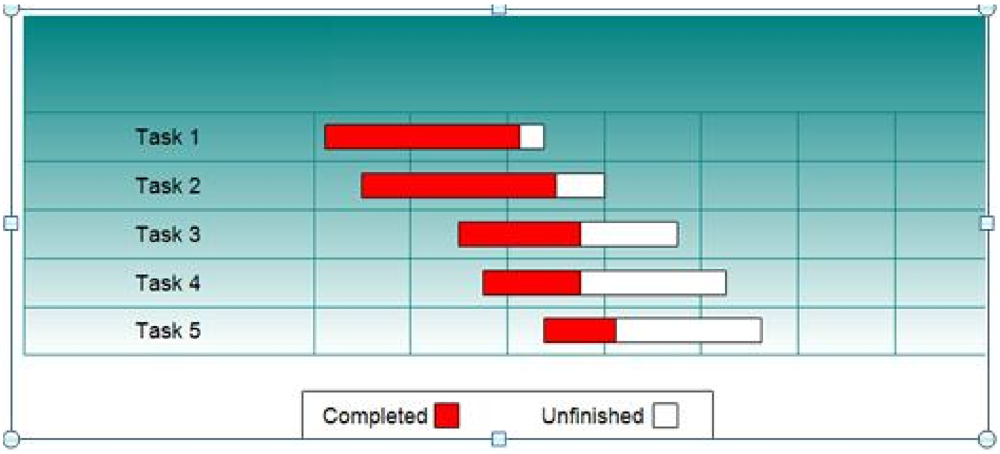


Figure 1: Task network

8.3.3 Timeline Chart

Activity	I week	II week	III week	IV week	V Week	VI week	VII week	VIII week	IX week
	Aug 4	Aug 11	Aug 18	Aug 25	Sept 1	Sept 8	Sept 15	Sept 22	Sept 29
Initiate the project									
Communication									
Literature survey									
Define scope									
Develop SRS									
Plan the project									
Design mathematical model									
Feasibility Analysis									
Develop work breakdown structure									
Planning project schedule									
Design UML and other diagrams									
Design test plan									
Design risk management plan									

Figure 2: Timeline Chart

8.3.4 Timeline Chart

Activity	XI week	XII week	XIII week	XIV week	XV week	XVI week	XVII week	XVIII week	XIX week	XX week	XXI week	XXII week
	Jan 5	Jan 15	Jan 19	Jan 26	Feb 2	Feb 9	Feb 16	Feb 23	Mar 2	Mar 9	Mar 16	April 25
Execute the project												
Build and test basic functional unit												
Build and test database with login and session maintenance facility												
Build and test Bluetooth mode												
Build and test security features												

Figure 3: Timeline Chart

8.4 Team Organization

8.5 Team structure

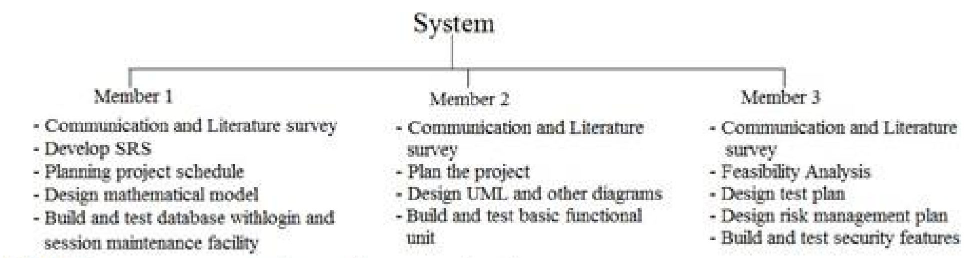


Figure 4: Team structure

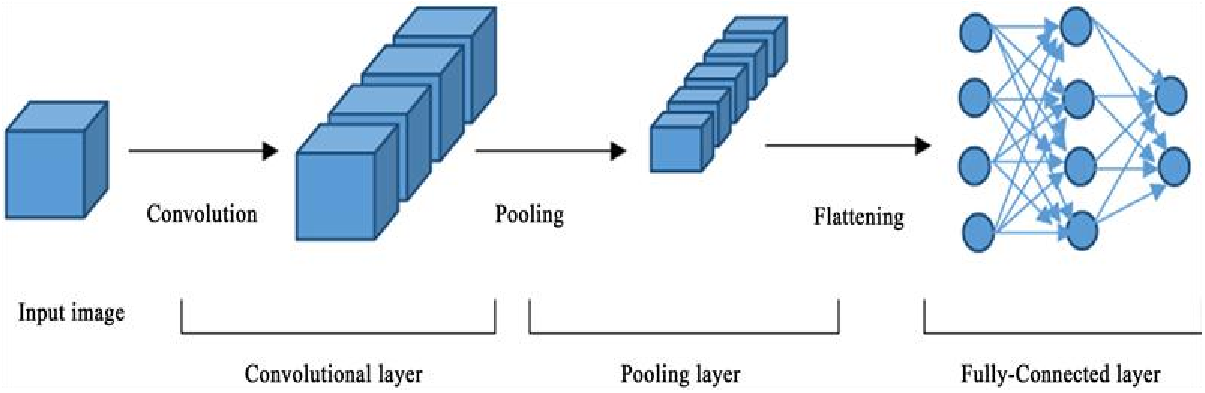
8.5.1 Management reporting and communication

Sr No.	Month	Description
1	June	Discussion with guide regarding domain. Searching for IEEE paper for domain.
2	July	Short listing of IEEE papers within domain. Selection of IEEE paper.
3	August	Deciding Project name. Submission of Synopsis.
4	September	Requirement analysis. Designing of models.
5	October	Report preparation. Stage-I report submission.

Table 2: Management reporting and communication

9 Project Design

9.1 CNN Module



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The Fully Connected (FC) layer consists of the weights and biases along with the neurons and is used to connect the neurons between two different layers. These layers are usually placed before the output layer and form the last few layers of a CNN Architecture.

9.2 Data Flow Diagrams

9.2.1 Level 0

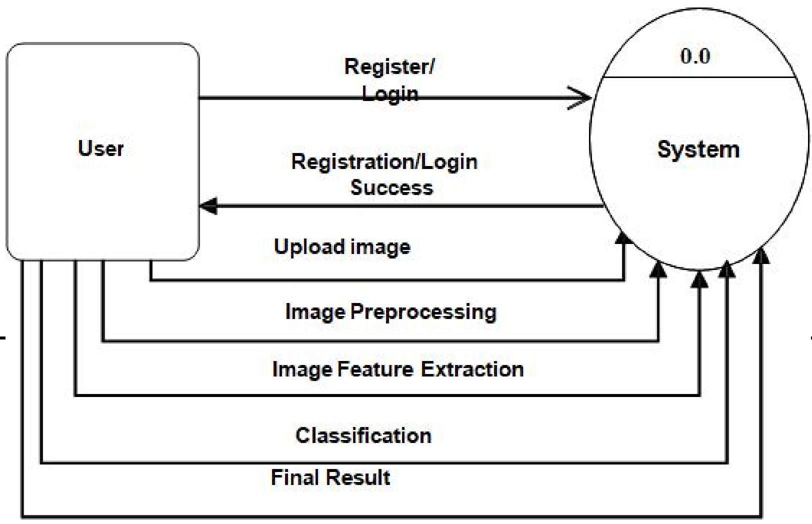


Figure 5: Data Flow Level 0

9.2.2 Level 1

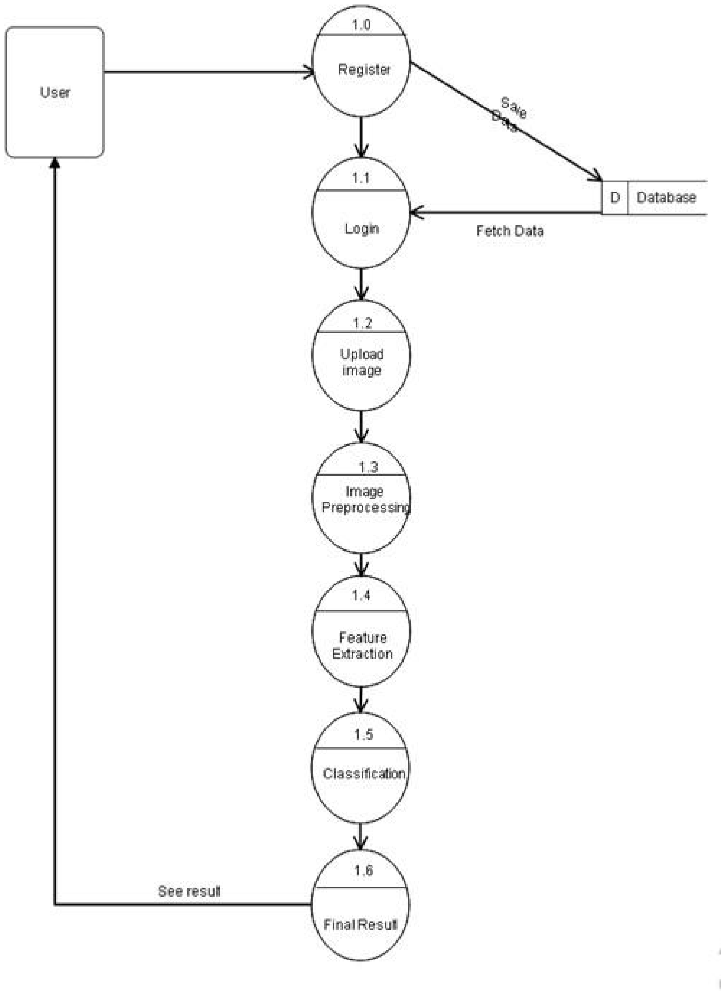


Figure 6: Data Flow Level 1

9.3 Use Case Diagram

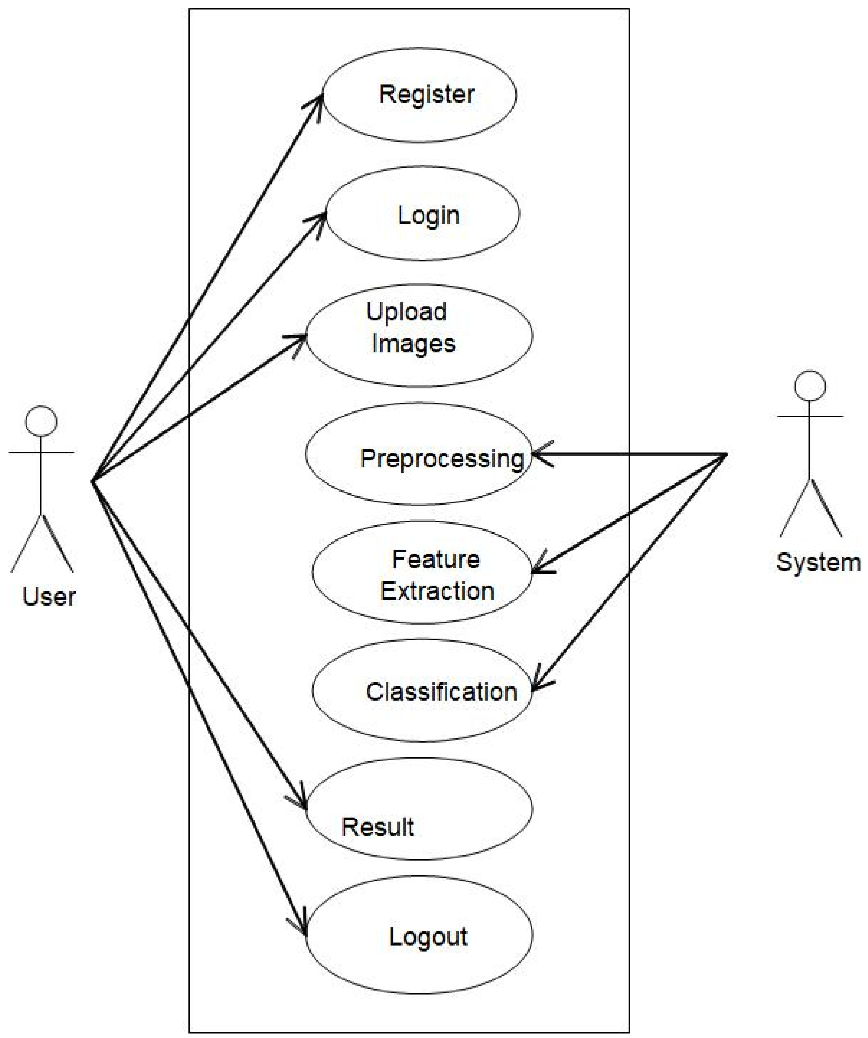


Figure 7: Use Case Diagram

9.4 Class Diagram

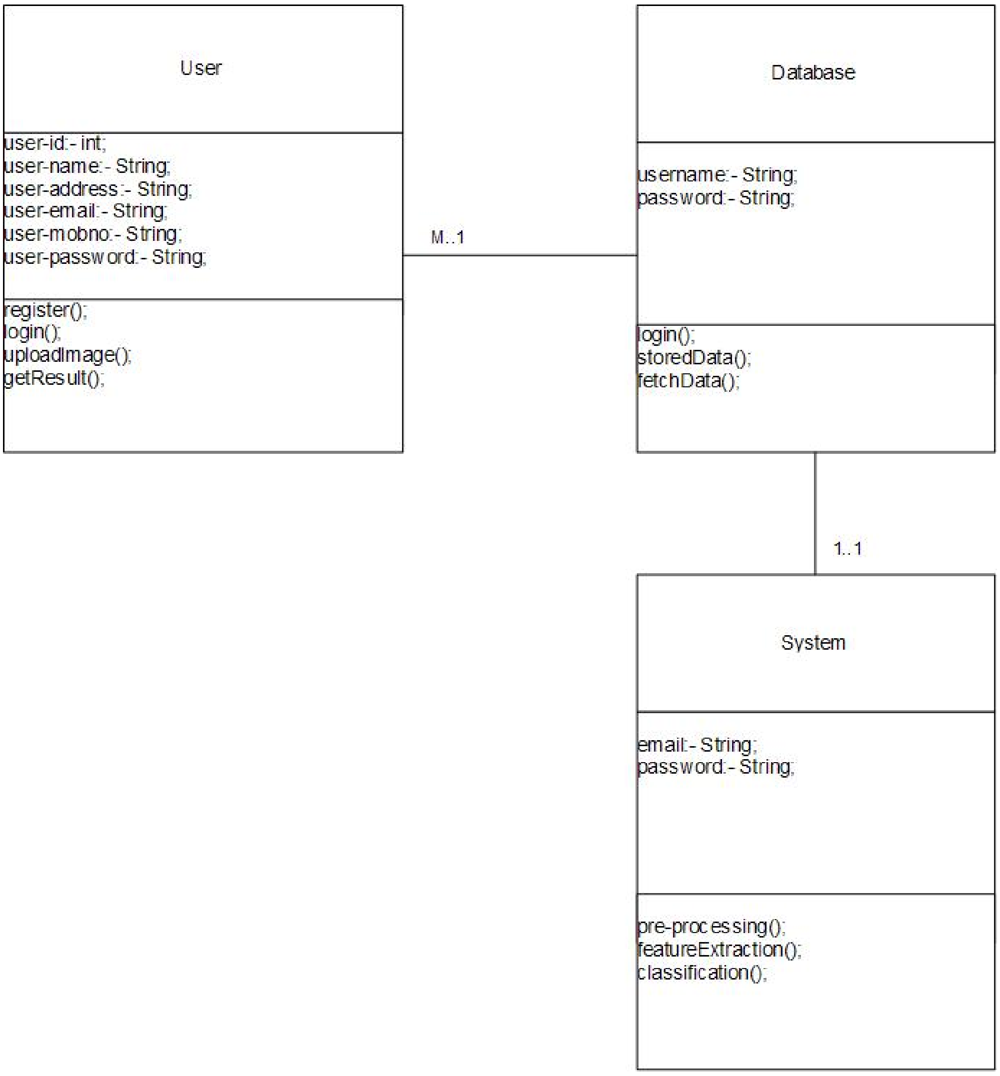


Figure 8: Class Diagram

9.5 Activity Diagram

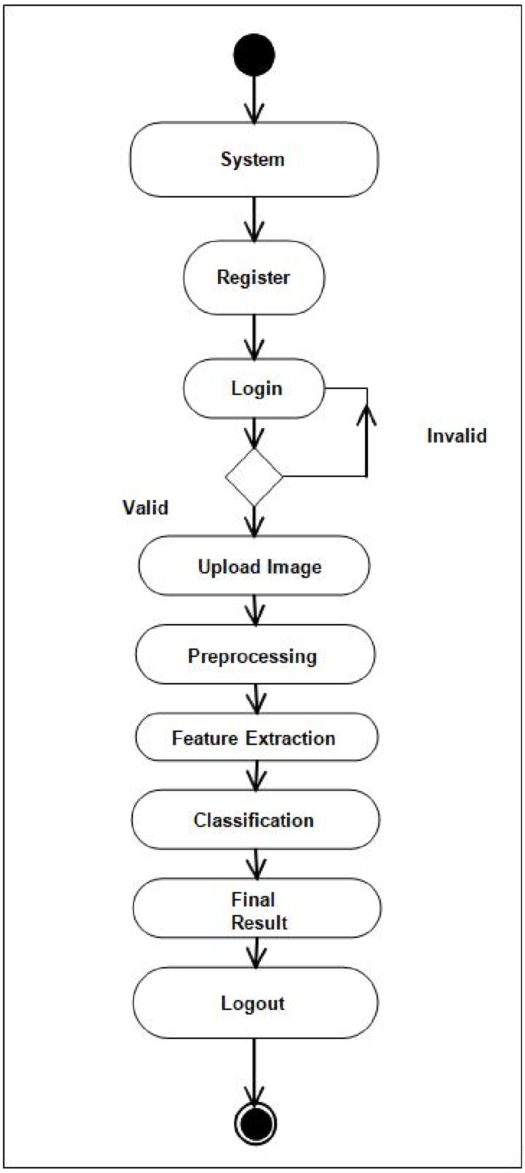


Figure 9: Activity Diagram

9.6 Sequence Diagram

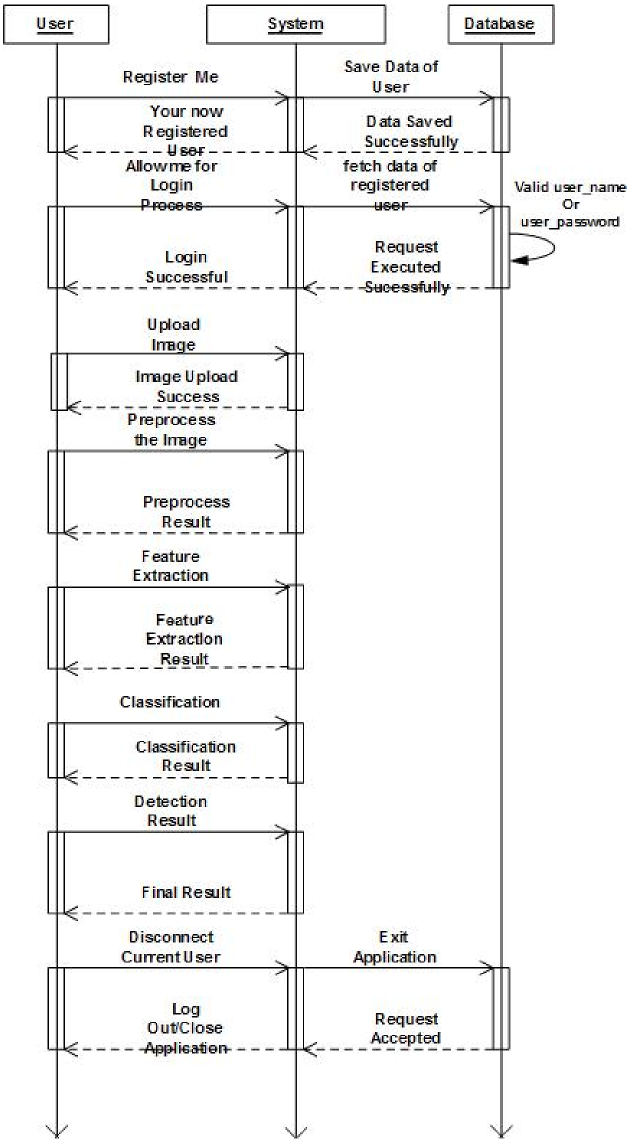


Figure 10: Sequence Diagram

9.7 Component Diagram

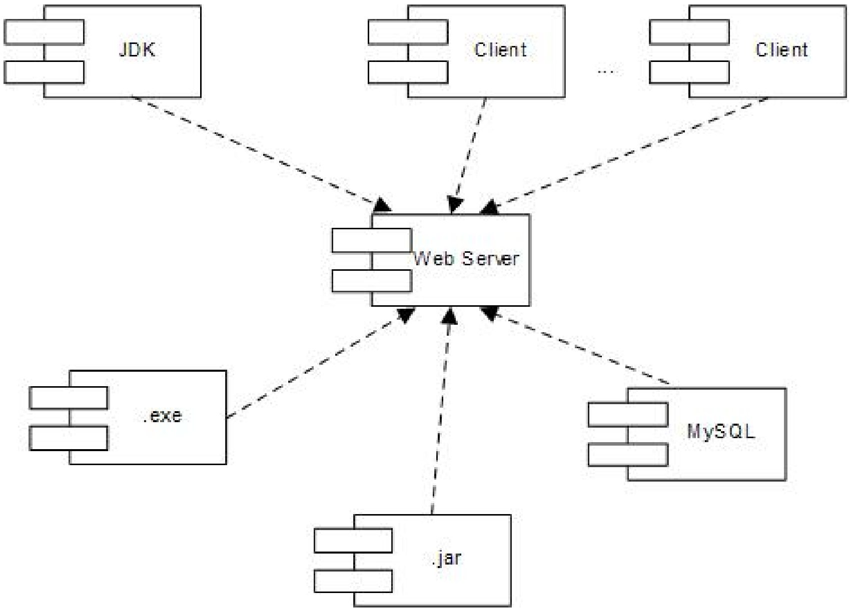


Figure 11: Component Diagram

9.8 Deployment Diagram

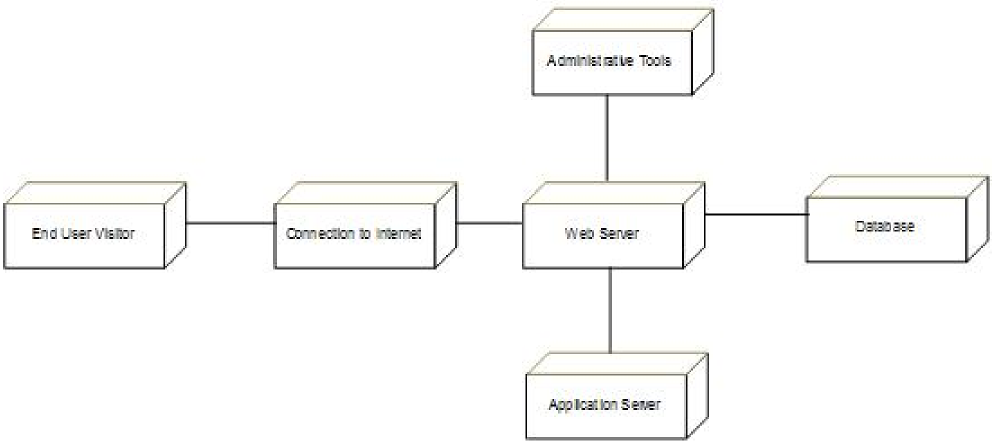


Figure 12: Deployment Diagram

10 Testing

This document is a procedural guide for listing the testing activities that should be carried out for the Keyword query routing project. It describes the software test environment for testing, identifies the tests to be performed, and provides schedules for test activities.

10.1 Purpose of the Document

The Purpose and objective is:

- Identify all the activities involved in testing,
- Resources required executing testing activities and monitoring mechanisms.
- To describe strategies for generating system test cases
- Improve the Reliability of system system
- Reduce the computation time and
- Maximize the security of private data.

10.2 Black Box Testing

Black box testing methods focus on the functional requirements in the software. That is, black box testing enables us to derive sets of input conditions that will fully exercise. All functional requirements of the program Black box testing attempts to find errors in the following categories:

- Incorrect or missing function
- Interface errors
- Errors in data structure or external job access
- Performance errors
- Initialization and termination errors.

In the proposed application with the help of this technique, we do not use the code to determinea test suite; rather, knowing the problem that weare trying to solve, we come up with four types of test data:

1. Easy-to-compute data,
2. Typical data,
3. Boundary / extreme data,
4. Bogus data.

But in our application we does not provide any external data, the role of user is only to give number of nodes for formation of clusters and for the formation of sink node.

10.3 White Box Testing

White box testing is a set case design method that uses the control structure of the procedural design to derive test cases. Using white box testing methods, we can derive test cases that:

- Guarantee that all independent paths within a module have been exercised at least once.
- Exercise all logical decisions on their true and false sides.
- Execute all loops at their boundaries and within their operational bounds.
- Exercise internal data structures to ensure their validity.

In the proposed application the white box testing is done by the developer implemented the code, the implements code is studied by the coder, determines all legal (valid and invalid) and illegal inputs and verifies the outputs against the expected outcomes, which is also determined by studying the implementation code.

10.4 Testing Types

10.4.1 Unit Testing

Unit testing enables a programmer to detect error in coding. A unit test focuses verification of the smallest unit of software design. This testing was carried out during the coding itself. In this testing step, each module going to be work satisfactorily as the expected output from the module.

Project Aspect

The front end design consists of various forms. They were tested for data acceptance. Similarly, the back-end also tested for successful acceptance and retrieval of data. The unit testing is done on the developed code. Mainly the unit testing is done on modules.

10.4.2 Integration Testing

Through each program work individually, they should work after linking together. This is referred to as interfacing. Data may be lost across the interface; one module can have adverse effect on the other subroutines after linking may not do the desired function expected by the main routine. Integration testing is the systematic technique for constructing the program structure while at the same time conducting test to uncover errors associated with the interface. Using integrated test plan prepared in the design phase of the system development as a guide, the integration test was carried out. All the errors found in the system were corrected for the next testing step.

10.4.3 System Testing

After performing the integration testing, the next step is output testing of the proposed system. No system could be useful if it does not produce the required output in a specified format. The outputs generated are displayed by the user. Here the output format is considered in to two ways. One in on screen and other in printed format.

10.4.4 System testing for the current system:

In this level of testing, testing the system after integrating developed sub module of the project and testing whether system is giving correct output or not. Developed sub module was integrated and the flow of information was checked. It was also checked that whether the flow of data is as per the requirements or not. It was also checked that whether any particular module is non-functioning or not i.e. once the integration is over each and every module is functioning in its entirety or not.
In this level of testing, testing can be done for following points: -

- Whether developed forms are properly working or not?
- Whether developed forms are properly linked or not?
- Whether images are properly displayed or not?
- Whether data retrieval is proper or not?

10.5 Test Plan

Test Plan Identifier: cost-sensitive
It is use to identify test plan uniquely.

1. Purpose of the Test Plan Document

   The main purpose of this document is to fit a particular project's needs. It documents and tracks the necessary information required to effectively define the approach to be used in the testing of the projects product. The Test Plan document is created during the Planning Phase of the project. Its intended audience is the project manager, project team, and testing team.

2. Objective of Test Panning

   To find as many defects as possible and get them fix.

3. Items to be Tested OR Not to be Tested

   Describe the items/features/functions to be tested that are within the scope of this test plan. Include a description of how they will be tested, when, by whom, and to what quality standards. Also include a description of those items agreed not to be tested.

4. Items to be tested

   - Overall functionality of the application.

- User Interface of the application.

5. Not to be Tested
   Performance of the application.

6. Test Approach

   Describe the overall testing approach to be used to test the projects product. Provide an outline of any planned tests. There are many approaches like:

   - Black Box Testing White Box Testing

   Here we used Black Box Testing approach. In Black Box Testing we just give input to the system and check its output without checking how system processes it.

7. Test Pass OR Test Fail Criteria

   When actual and expected results are same then test will be passed. When actual and expected results are different then test will be failed.

8. Test Entry OR Exit Criteria

   Describe the entry and exit criteria used to start testing and determine when to stop testing.
   - Entry criteria: As soon as we have requirement we can start testing.
   - Exit criteria: When bug rate fall below certain level we can stop testing.

9. Test Suspension OR Resumption Criteria

   Describe the suspension criteria that may be used to suspend all or portions of testing. Also describe the resumption criteria that may be used to resume testing.
   - Suspension criteria: If there is large change in application like change in requirements we can suspend work for some time.
   - Resumption criteria: After resolving the respective problem we can resume work. item Testing Type

   It describes which testing types we are going to follow in our testing lifecycle. Here we are using:
   - Black Box Testing
   - Functional Testing
   - UI Testing
   - Integration Testing

10. Test Cases

    Test cases are built around specifications and requirements, i.e., what the application is supposed to do. Test cases are generally derived from external

descriptions of the software, including specifications, requirements and design parameters. Although the tests used are primarily functional in nature, non-functional tests may also be used. The test designer selects both valid and invalid inputs and determines the correct output without any knowledge of the test object's internal structure. Test Design Techniques
Typical black-box test design techniques include:

- Decision table testing
- All-pairs testing
- State transition Analysis
- Equivalence partitioning
- Boundary value analysis
- Cause–effect graph
- Error guessing

Advantages

- Efficient when used on large systems.
- Since the tester and developer are independent of each other, testing is balanced and unprejudiced.
- Tester can be non-technical.
- There is no need for the tester to have detailed functional knowledge of system.
- Tests will be done from an end user's point of view, because the end user should accept the system. (This testing technique is sometimes also called Acceptance testing.)
- Testing helps to identify vagueness and contradictions in functional specifications.
- Test cases can be designed as soon as the functional specifications are complete.

Disadvantages

- Test cases are challenging to design without having clear functional specifications.
- It is difficult to identify tricky inputs if the test cases are not developed based on specifications.
- It is difficult to identify all possible inputs in limited testing time. As a result, writing test cases may be slow and difficult.
- There are chances of having unidentified paths during the testing process.
- There is a high probability of repeating tests already performed by the programmer.

1] Test case For Registration Page:

Table 3: Test cases For Registration Page

Test Case ID	Test Case Procedure	Input Data	Expected Output	Actual Output	Test Status
1	Checking the functionality of Submit Button	1.Enter valid User name 2.Enter valid email-id 3.Enter valid Password 4.Enter valid date 5.Enter valid contact no. 6.Click on Register Button	Accept & Login page should be displayed	Accept & Login page displayed	Pass
2	Checking the functionality of Submit Button	1.Enter invalid User name 2. Enter valid email-id 3. Enter valid Password 4. Enter valid date 5. Enter valid contact no. 6. Click on Register Button	Error should come & Register page should be displayed	Error comes & Register page displayed	Pass
3	Checking the functionality of Submit Button	1.Enter valid User name 2. Enter invalid email-id 3. Enter valid Password 4. Enter valid date 5. Enter valid contact no. 6. Click on Register Button	Error should come & Register page should be displayed	Error comes & Register page displayed	Pass

4	Checking the functionality of Submit Button	1.Enter valid User name 2. Enter valid email-id 3. Enter invalid Password 4. Enter valid date 5. Enter valid contact no. 6. Click on Register Button	Error should come & Register page should be displayed	Error comes & Register page displayed	Pass
5	Checking the functionality of Submit Button	1.Enter valid User name 2. Enter valid email-id 3. Enter valid Password 4. Enter invalid date 5. Enter valid contact no. 6. Click on Register Button	Error should come & Register page should be displayed	Error comes & Register page displayed	Pass
6	Checking the functionality of Submit Button	1.Enter valid User name 2. Enter valid email-id 3. Enter valid Password 4. Enter valid date 5. Enter invalid contact no. 6. Click on Register Button	Error should come & Register page should be displayed	Error comes & Register page displayed	Pass

2] Test case For Login Page:

Table 4: Test cases For Login Page

Test Case ID	**Test Case Procedure**	**Input Data**	**Expected Output**	**Actual Output**	**Test Status**

1	Checking the functionality of Login Button	1.Enter valid User name 2 .Enter valid Password 3. Click on Login Button	Accept & User's home page should be display	Accept & User's home page displayed	Pass
2	Checking the functionality of Login Button	1.Enter valid User name 2 .Enter Invalid Password 3. Click on Login Button	Error should come & Login page should be displayed	Error should come & Login page displayed	Pass
3	Checking the functionality of Login Button	1.Enter Invalid User name 2 .Enter valid Password 3. Click on Login Button	Error should come & Login page should be displayed	Error should come & Login page displayed	Pass
4	Checking the functionality of Login Button	1.Enter Invalid User name 2 .Enter Invalid Password 3. Click on Login Button	Error should come & Login page should be displayed	Error should come & Login page displayed	Pass

3] Test case For Upload Leaf Image :

Table 5: Test cases For Extract Real-time Tweets

Test Case ID	**Test Case Procedure**	**Input Data**	**Expected Output**	**Actual Output**	**Test Status**
1	Checking the functionality of upload button	1. Passing the valid file format.	preprocess image	Display all features in view page	Pass
2	Checking the functionality of upload button	Passing the valid file format	Display Error message on home Page	Display Error message on home page	Pass

4] Test case For Disease Classification:

Table 6: Test cases For Disease Classification

Test Case ID	Test Case Procedure	Input Data	Expected Output	Actual Output	Test Status
1	Checking the functionality of Classify button	1. Passing the valid leaf dataset.	Display all features with classification result on Result page.	Display all features with classification result on Result page.	Pass
2	Checking the functionality of Classify button	1. Passing the invalid leaf dataset.	Display Error message on home page	Display Error message on home Page	Pass

11 Code

```
import os
import cv2
import pandas as pd
import numpy as np
import tensorflow as tf
import matplotlib.pyplot as plt

import warnings
warnings.filterwarnings("ignore")
```

```
# File Directory for both the train and test
train_path = "Data/train"
val_path = "Data/valid"
test_path = "Data/test"
```

```
def GetDatasetSize(path):
    num_of_image = {}
    for folder in os.listdir(path):
        # Counting the Number of Files in the Folder
        num_of_image[folder] = len(os.listdir(os.path.join(path, folder)));
    return num_of_image;

train_set = GetDatasetSize(train_path)
val_set = GetDatasetSize(val_path)
test_set = GetDatasetSize(test_path)
print(train_set,"\n\n",val_set,"\n\n",test_set)
```

```
labels = ['adenocarcinoma', 'large.cell.carcinoma', 'normal','squamous.cell.carcinoma' ]
# labels = ['squamous.cell.carcinoma', 'normal', 'adenocarcinoma', 'large.cell.carcinoma']
train_list = list(train_set.values())
val_list = list(val_set.values())
test_list = list(test_set.values())

x = np.arange(len(labels))  # the label locations
width = 0.25  # the width of the bars
```

```
fig, ax = plt.subplots()
rects1 = ax.bar(x - width, train_list, width, label='Train')
rects2 = ax.bar(x, val_list, width, label='Val')
rects3 = ax.bar(x + width, test_list, width, label='Test')
```

```
# Add some text for labels, title and custom x-axis tick labels, etc.
ax.set_ylabel('Images Count')
ax.set_title('Dataset')
# ax.set_xticks(x, labels)
ax.set_xticks(x)
ax.set_xticklabels(labels)
```

```
plt.xticks(rotation=15)
ax.legend()
```

```
ax.bar_label(rects1)
ax.bar_label(rects2)
ax.bar_label(rects3)
```

```
fig.tight_layout()
```

```
plt.show()
```

```
import tensorflow.keras
from tensorflow.keras import layers
from tensorflow.keras import Model
from tensorflow.keras.models import Sequential
from tensorflow.keras.preprocessing import image
from tensorflow.keras.callbacks import ModelCheckpoint, EarlyStopping
from tensorflow.keras.preprocessing.image import ImageDataGenerator
from tensorflow.keras.applications.vgg16 import VGG16, preprocess_input
from tensorflow.keras.models import load_model, Model
from tensorflow.keras.layers import Dense,Conv2D, Flatten, MaxPool2D, Dropout
```

```
train_datagen = ImageDataGenerator(rescale = 1.0/255.0,
                                  horizontal_flip = True,
                                  fill_mode = 'nearest',
                                  zoom_range=0.2,
                                  shear_range = 0.2,
                                  width_shift_range=0.2,
                                  height_shift_range=0.2,
                                  rotation_range=0.4)

train_data = train_datagen.flow_from_directory(train_path,
                                                   batch_size = 5,
                                                   target_size = (64,64),
                                                   class_mode = 'categorical')
```

```
val_datagen = ImageDataGenerator(rescale = 1.0/255.0)
val_data = val_datagen.flow_from_directory(val_path,
                                                   batch_size = 5,
                                                   target_size = (64,64),
                                                   class_mode = 'categorical')
```

```
model = Sequential()

# Convolutional Layer with input shape (224,224,3)
model.add(Conv2D(filters=32, kernel_size= (3,3), activation= 'relu', input_shape=(64,64,3)) )
```

```
model.add(Conv2D(filters=32, kernel_size=(3,3), activation='relu' ))
model.add(MaxPool2D(pool_size=(2,2)))
```

```
model.add(Conv2D(filters=64, kernel_size=(3,3), activation='relu' ))
model.add(MaxPool2D(pool_size=(2,2)))
```

```
model.add(Conv2D(filters=128, kernel_size=(3,3), activation='relu' ))
model.add(MaxPool2D(pool_size=(2,2)))
```

```
model.add(Dropout(rate=0.25))
```

```
model.add(Flatten())
model.add(Dense(units=64, activation='relu'))
model.add(Dropout(rate=0.25))
model.add(Dense(units=4, activation='sigmoid'))
```

```
model.compile(optimizer='adam', loss='categorical_crossentropy', metrics=['accuracy']  )

model.summary()
```

```
# Adding Model check point Callback

mc = ModelCheckpoint(
    filepath="./ct_cnn_best_model.hdf5",
    monitor= 'val_accuracy',
    verbose= 1,
    save_best_only= True,
    mode = 'auto'
    );
```

```
call_back = [mc];
```

```
# Fitting the Model
cnn = model.fit(
    train_data,
    steps_per_epoch = train_data.samples//train_data.batch_size,
    epochs = 32,
    validation_data = val_data,
    validation_steps = val_data.samples//val_data.batch_size,
    callbacks = call_back
    )
```

```
# Loading the Best Fit Model
model = load_model("./ct_cnn_best_model.hdf5")
```

```
# Checking the Accuracy of the Model
accuracy_cnn = model.evaluate_generator(generator= test_data)[1]
print(f"The accuracy of the model is = {accuracy_cnn*100} %")
```

```
# Plot model performance
acc = cnn.history['accuracy']
val_acc = cnn.history['val_accuracy']
loss = cnn.history['loss']
val_loss = cnn.history['val_loss']
epochs_range = range(1, len(cnn.epoch) + 1)

plt.figure(figsize=(15,5))
```

```
plt.subplot(1, 2, 1)
plt.plot(epochs_range, acc, label='Train Set')
plt.plot(epochs_range, val_acc, label='Val Set')
plt.legend(loc="best")
plt.xlabel('Epochs')
plt.ylabel('Accuracy')
plt.title('Model Accuracy')
```

```
plt.subplot(1, 2, 2)
plt.plot(epochs_range, loss, label='Train Set')
plt.plot(epochs_range, val_loss, label='Val Set')
plt.legend(loc="best")
plt.xlabel('Epochs')
plt.ylabel('Loss')
plt.title('Model Loss')
```

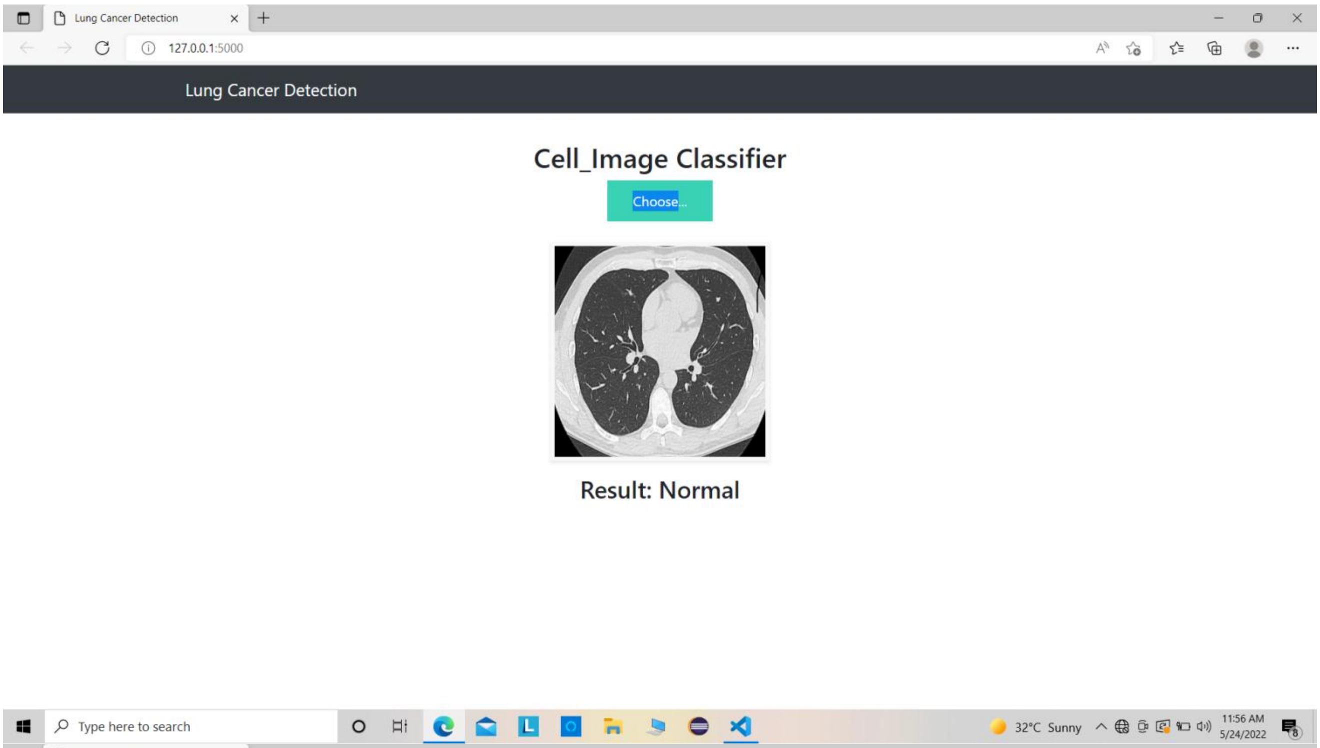
```
plt.tight_layout()
plt.show()
```

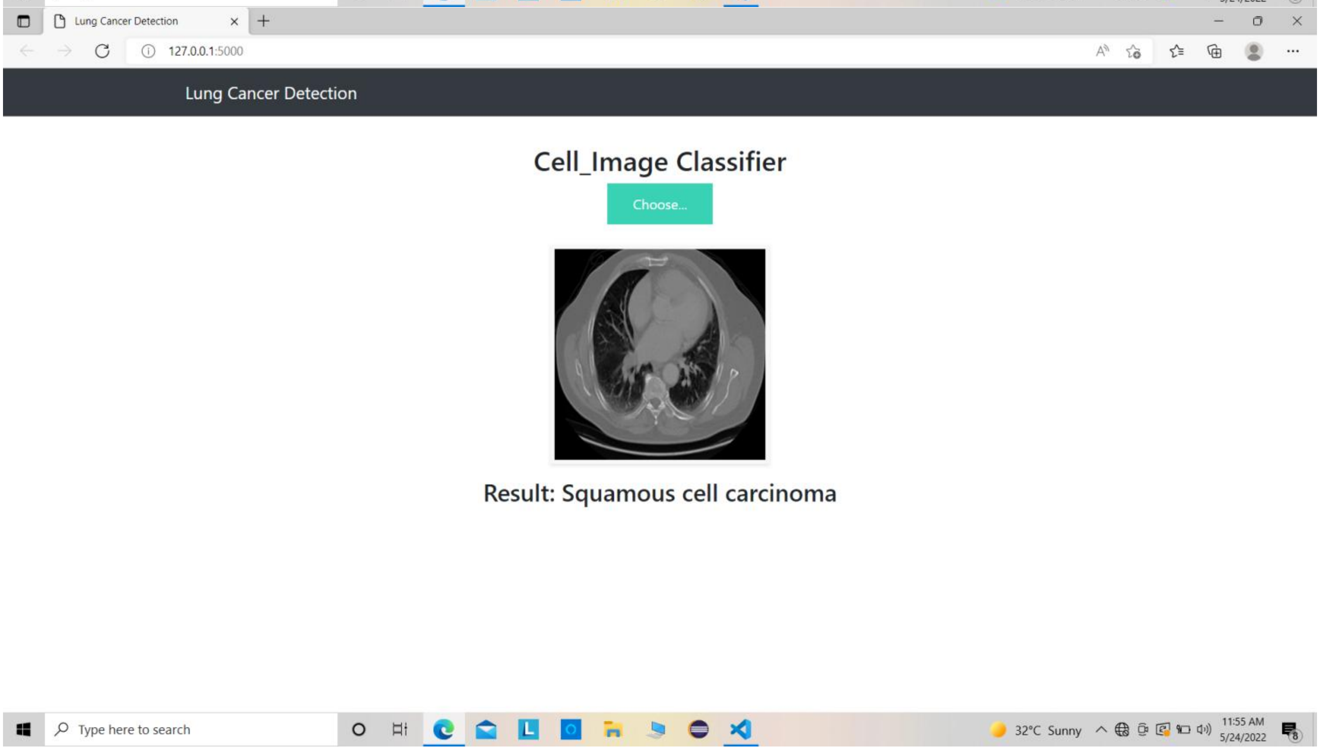
```python
algos = ['CNN']
accuracy = [accuracy_cnn]
accuracy = np.floor([i * 100 for i in accuracy])

fig = plt.figure(figsize = (10, 5))

# creating the bar plot
plt.bar(algos, accuracy, color ='green', width = 0.3)

plt.xlabel("Algorithms Applied")
plt.ylabel("Accuracy")
plt.show()
```

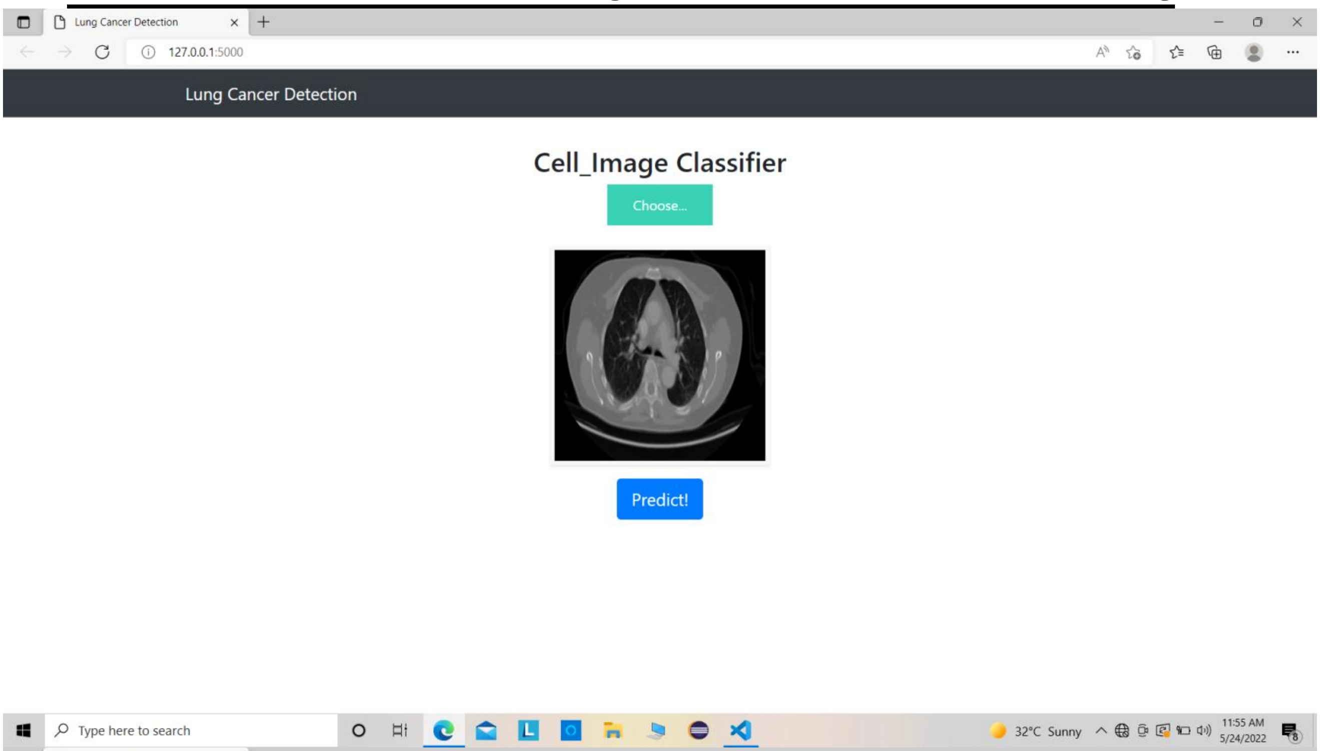
```python
def chestScanPrediction(path, _model):
    classes_dir = ["Adenocarcinoma","Large cell carcinoma","Normal","Squamous cell carcinoma"]
    # Loading Image
    img = image.load_img(path, target_size=(64,64))
    # Normalizing Image
    norm_img = image.img_to_array(img)/255
    # Converting Image to Numpy Array
    input_arr_img = np.array([norm_img])
    # Getting Predictions
    pred = np.argmax(_model.predict(input_arr_img))
    # Printing Model Prediction
    print("The Patient Cancer is",classes_dir[pred])
```

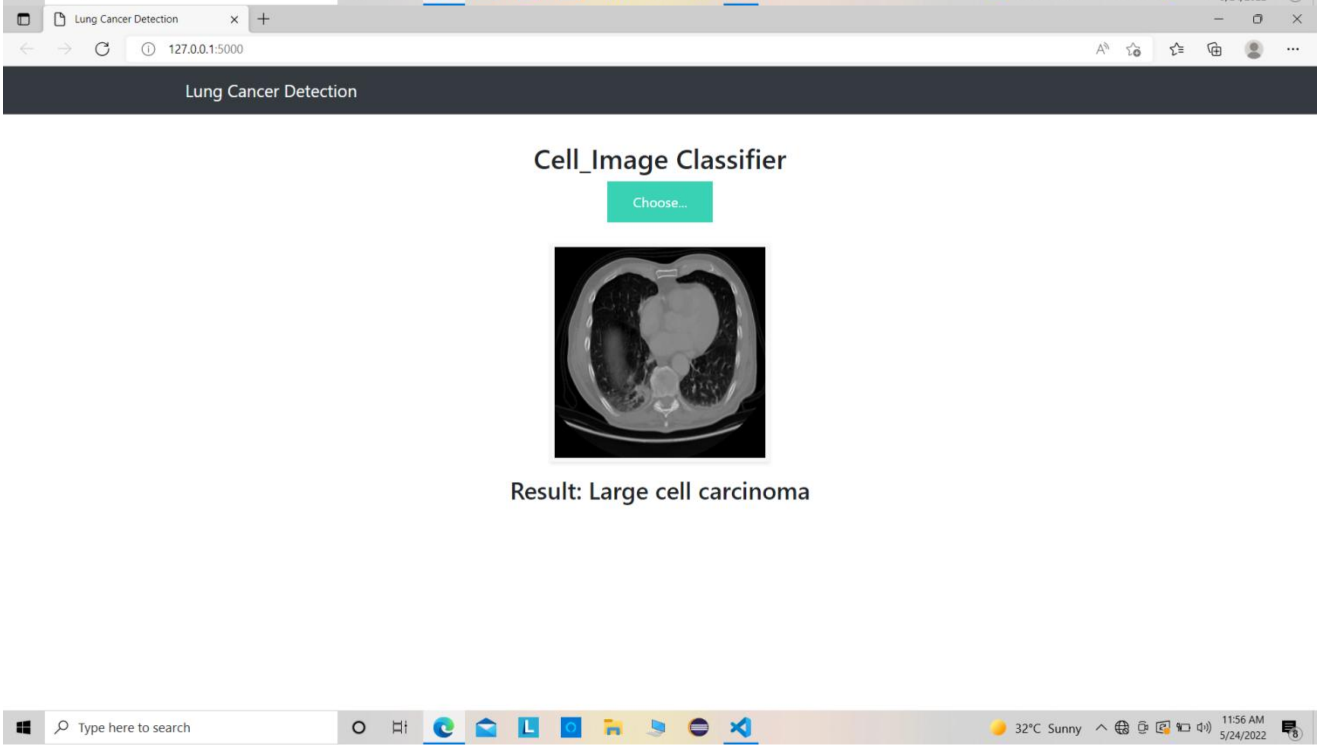
```python
path = r"C:\Final Project\Lung Cancer Final\ml\Data\test\squamous.cell.carcinoma\000108 (6).png"
chestScanPrediction(path,model)
```

12 Experimental Results






Lung Cancer Detection
127.0.0.1:5000
Lung Cancer Detection
Cell_Image Classifier
Choose...
Predict!
Type here to search
32°C Sunny
11:55 AM
5/24/2022


Lung Cancer Detection
127.0.0.1:5000
Lung Cancer Detection
Cell_Image Classifier
Choose...
Result: Large cell carcinoma
Type here to search
32°C Sunny
11:56 AM
5/24/2022

13 Conclusion

In this paper we proposed the fully automatic deep learning based method for detection of lung cancer in whole slide histopathology images. VGG16 and ResNet50 CNN architectures were compared and the first one shows higher AUC and patch classificiation accuracy. Presented results shows that convolutional neural networks have potential to perform lung cancer diagnose from whole slide images, but more effort is needed to increase classification accuracy. In future work next steps will be increasing the training set size, adding image augmentation and stain normalization. Also, we will try training from the scratch instead of using weights pretrained on ImageNet.

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